Final Project Report
Financial Applications of Blockchains and Distributed Ledgers

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1 Introduction

Automated market makers (AMMs) have become a central venue for on-chain liquidity. Unlike centralised limit-order books, where active market makers compete on quotes, AMMs rely on passive liquidity deposited by users across price ranges. Uniswap V3 extends this model with *concentrated liquidity*: each liquidity provider (LP) chooses a tick interval and earns fees only while the pool price remains inside that range. Execution quality for takers and the risk–return profile for LPs therefore depend not only on how much capital is in the pool, but on *where* that capital sits relative to the current price.

This report presents a six-month empirical study of one of the deepest on-chain USDC/WETH venues. All analysis targets the Uniswap V3 pool at contract `0x88e6A0c2dDD26FEEb64F039a2c41296FcB3f5640` (0.05% fee tier, tick spacing 10) on Ethereum mainnet, with token0 USDC (6 decimals) and token1 WETH (18 decimals). The swap study window runs from 1st October 2025 to 31st March 2026; liquidity-changing events are replayed from deployment so that daily pool states can be reconstructed consistently. The overarching objective is to link the liquidity distribution of this pool to taker execution costs and LP economics, and to assess whether impermanent loss (IL) can be mitigated with a delta hedge on Hyperliquid ETH perpetual futures. Data are extracted directly from an Ethereum archive node via RPC; off-chain hedge inputs come from the Hyperliquid API. The accompanying codebase reproduces every table and figure from documented scripts.

The project follows the required five-module structure. Module 1 builds and validates the on-chain dataset: swap, Mint/Burn, and Collect logs, daily `slot0` prices, and tick-level liquidity maps. Module 2 characterises how liquidity is distributed across prices and how that landscape evolves, including TVL decomposition and concentration metrics (in-range liquidity ratio and liquidity Herfindahl Hirschman index). Module 3 implements a Uniswap V3 swap simulator on each daily snapshot, maps price impact over trade size, and compares simulated costs with effective spreads from observed swaps. Module 4 shifts to the LP side: five synthetic $100,000 positions with different range widths, with fee income, impermanent loss, and net P&L decomposed over the window. Module 5 treats the LP payoff as a short-gamma exposure, derives delta and gamma analytically, and backtests discrete delta hedging with hourly Hyperliquid perp and funding data. The remainder of the report is organised accordingly; an appendix provides supplementary liquidity-profile figures over the full price range.

2 Module 1 — On-Chain Data Extraction

Module 1 builds the raw on-chain data layer used by all later analysis on the pool mentioned in the introduction. The swap study window is from the 1st of October 2025 to the 31st of March 2026, while liquidity-changing events are read from the pool deployment block, 12,376,729, so that the opening liquidity map can be reconstructed from the full history of Mint and Burn events.

2.1 Extracted Files

All event data was extracted directly using Ethereum archive-node RPC calls, with the dedicated extraction script resolving the study-window blocks from UTC timestamps, downloading raw pool logs, decoding them with the pool ABI, resolving timestamps once per unique block, and writing them to compact Parquet files. All four required files are reproduced from scratch by a single documented entry point, `src/module1/data_extraction.py`, which takes as inputs an archive RPC endpoint URL and the desired pool address (`data_extraction.py extract all`). The output data dictionary can be found in the repository's README.

The main swap file, `swap_events.parquet`, contains 1,095,286 decoded swaps across 637,982 unique blocks. Their precise timestamp range is 2025-10-01 00:00:11 UTC to 2026-03-31 23:59:59

Table 1: Volume cross-check on `swap_events.parquet` (1 Oct 2025 – 31 Mar 2026).

Direction	Swaps	Absolute USDC notional (USD bn)
`buy_weth`	534,929	10.702
`sell_weth`	560,357	10.690
Total	1,095,286	21.392

UTC meaning the file data is bounded by the required six-month window. The underlying cumulative swap volume is nearly balanced, with 534,929 swaps as `buy_weth` and 560,357 as `sell_weth`. Aggregating absolute USDC notional gives a total pool volume of $21.39 billion USDC over the sample, split almost evenly between buys ($10.70 billion) and sells ($10.69 billion). This allows us to compare with the decoded trade direction convention and gives no indication of missing sign flips or decimal-scaling errors.

In the liquidity event file, `mint_burn_events.parquet`, there are 427,934 events, from contract deployment through to the end of the study window, with 204,398 Mints and 223,536 Burns. The support file `collect_events.parquet` contains 218,219 Collect events over the same historical range and is used later for fee analytics. The daily state files contain 182 midnight-UTC snapshots, where `slot0_snapshots.parquet` stores the pool price and active tick, while `liquidity_snapshots.parquet` stores 264,149 the initialized-tick rows, averaging around 1,451 ticks per snapshot.

2.2 Validation

We first apply schema and coverage checks to every generated file. These checks verify the required columns, reject duplicate event identities (`tx_hash`, `log_index`), require valid tick spacing and Mint/Burn signs thereby confirming that both slot0 and liquidity snapshots cover exactly 182 daily blocks. All checks pass on the submitted files.

The volume cross-check aggregates decoded swaps by direction and compares the resulting totals against the event-level signs. Since USDC is token0, a `buy_weth` swap has positive USDC flow into the pool whereas a `sell_weth` swap has positive WETH flow into the pool. Therefore the aggregate absolute USDC amount equals the reported USD notional total, $21.39 billion, and the buy/sell notional split differs by less than 0.2%. Table 1 reports the cross-check by direction with the buy/sell notional split differing by only 0.061%, and the aggregate absolute USDC amount equaling the reported USD notional total to the displayed precision.

We also compare each daily slot0 price with the most recent decoded swap price at or before the snapshot block. The validation table contains 181 comparable snapshots, 180 of these being within the 1 bp tolerance used by the script. The median absolute difference is 0.0000 bps, and the 95th percentile is 0.0018 bps. The only larger row is 11.71 bps on the snapshot from 2026-02-05, where the comparison swap is in the same block as the snapshot. This appears to be a same-block ordering edge case in the diagnostic table rather than a systematic price conversion problem: the remaining comparisons are effectively exact. Table 2 summarizes the full check, saved to `data/results/module_1/slot0_swap_price_check.parquet`.

Finally, the reconstructed liquidity map is checked against direct archive `pool.ticks()` calls. At snapshot block 23,479,243, we sampled 10 initialized ticks from `liquidity_snapshots.parquet` and queried the pool contract at that same historical block. Both `liquidityGross` and `liquidityNet` matched exactly for all 10 sampled ticks. This confirms that replaying the Mint and Burn history reconstructs the on-chain tick state at the snapshot block, with no discrepancies found in the spot-check. Table 3 lists the sampled ticks; the persisted artifact `data/results/module_1/liquidity_tick_spot_check.parquet` contains the full reconstructed and on-chain values for both `liquidityGross` and `liquidityNet`, which are bit-for-bit equal for every sampled tick.

Table 2: Daily slot0 price vs. last decoded swap price before the snapshot block.

Statistic	Value
Comparable snapshots	181
Within 1 bp tolerance	180
Median absolute difference (bps)	0.0000
95th percentile absolute diff. (bps)	0.0018
Maximum absolute difference (bps)	11.71
Snapshot of the maximum	2026-02-05 (same block)

Table 3: Reconstructed tick state vs. direct archive `pool.ticks()` at block 23,479,243. Values shown to 4 significant figures; the persisted artifact stores exact integers and all rows match exactly on both `liquidityGross` and `liquidityNet`.

Tick	L_{gross}	L_{net}	On-chain match
−887,270	2.308×10^{16}	$+2.308\times10^{16}$	Yes
−92,110	3.983×10^{11}	$+3.983\times10^{11}$	Yes
−76,280	2.965×10^{19}	$+2.965\times10^{19}$	Yes
−75,980	2.965×10^{19}	-2.965×10^{19}	Yes
0	1.357×10^{13}	$+1.357\times10^{13}$	Yes
100	100	+100	Yes
110	100	−100	Yes
100,000	1.357×10^{13}	-1.357×10^{13}	Yes
108,340	2.267×10^{9}	$+2.267\times10^{9}$	Yes
108,360	2.274×10^{9}	$+2.274\times10^{9}$	Yes

3 Module 2 — Liquidity Distribution Analysis

Module 2 studies how liquidity is distributed across prices in our chosen Uniswap V3 USDC/WETH 0.05% pool, and how this distribution changes over the 182 daily snapshots from 1st of October 2025 to 31st of March 2026. The analysis uses the tick-level liquidity maps and the matching `slot0` prices reconstructed in Module 1.

3.1 Task 2.1 — Liquidity Profile Visualisation

For each initialized tick, we convert the tick index into a human-readable price in USDC per WETH,

$$P(\tau) = \frac{10^{12}}{1.0001^{\tau}},$$

where the factor 10^{12} adjusts for the 6 decimals of USDC and 18 decimals of WETH. The plotted liquidity profile is the active liquidity after crossing each initialized tick. Figure 1 shows three representative snapshots: the start of the sample, on the 6th of February 2026 and the end of the sample. We use the 6th of February 2026 as the high-volatility snapshot because it is the lowest ETH price and lowest TVL day in our sample.

These graphs show the liquidity distribution at around 20% above and below the current price, as showing the liquidty distribution across the entire price range, gives us a lower resolution landscape. We will therefore use this graph for the rest of the analysis, however the full-range version is shown Figure 17 in Appendix A.

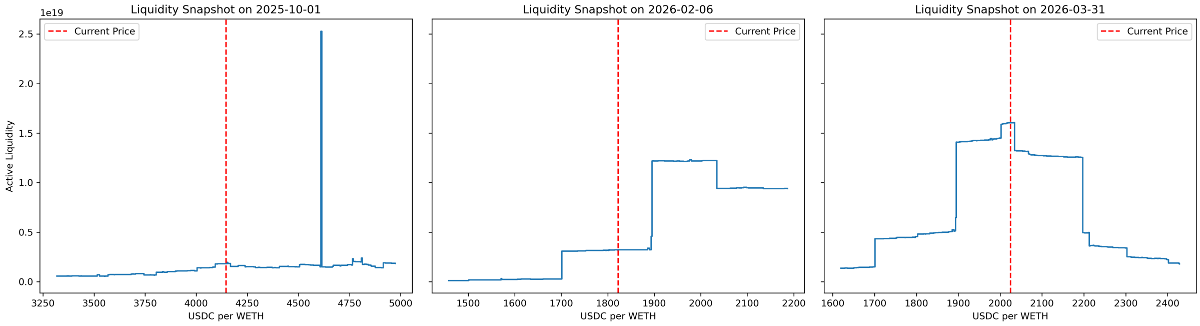

Figure 1: Fig 2.1 — Liquidity profiles around the current pool price for three snapshots. The red dashed line marks the current USDC/WETH price.

Our profiles show that liquidity is not uniformly distributed across prices. At the beginning of the window, when ETH traded around $4,146, most active liquidity was positioned below the current price. Remarkably, after the price decline into February and March, liquidity became much more concentrated around the realized trading range near $2,000. The monthly heatmaps in Figure 2 confirm this migration with late-sample liquidity being denser and closer to the price path than in October–December.

The graphs below show the liquidity distribution for each snapshot, using colour for concentration of liquidity, with the white line representing the current price. It is a tiled time-series graph, as it shows the heatmap for each day, in monthly groups. With this representation we can notice the migration of the liquidity distribution towards the current price, thanks to the change in colour of the dark sections at the top and bottom of the graph. There is also a clear concentration of liquidity around 2000$ throughout our sample. The persistent band around $2,000 appears to be a genuine liquidity concentration rather than a plotting artifact: it is present across multiple snapshots and is consistent with the underlying initialized-tick liquidity data. The persistent liquidity band around $2,000 likely reflects passive range-order liquidity: while out of range it earns no fees, but it sits as USDC ready to buy WETH if price falls into that level. This is consistent with LPs treating $2,000 as a major support/accumulation zone rather than continuously active market-making liquidity.

This graph also shows the liquidity distribution between roughly $1,500 and $5,000. In the appendix, the full price distribution graph using a Figure 18 in Appendix A shows the same distribution on a log price scale.

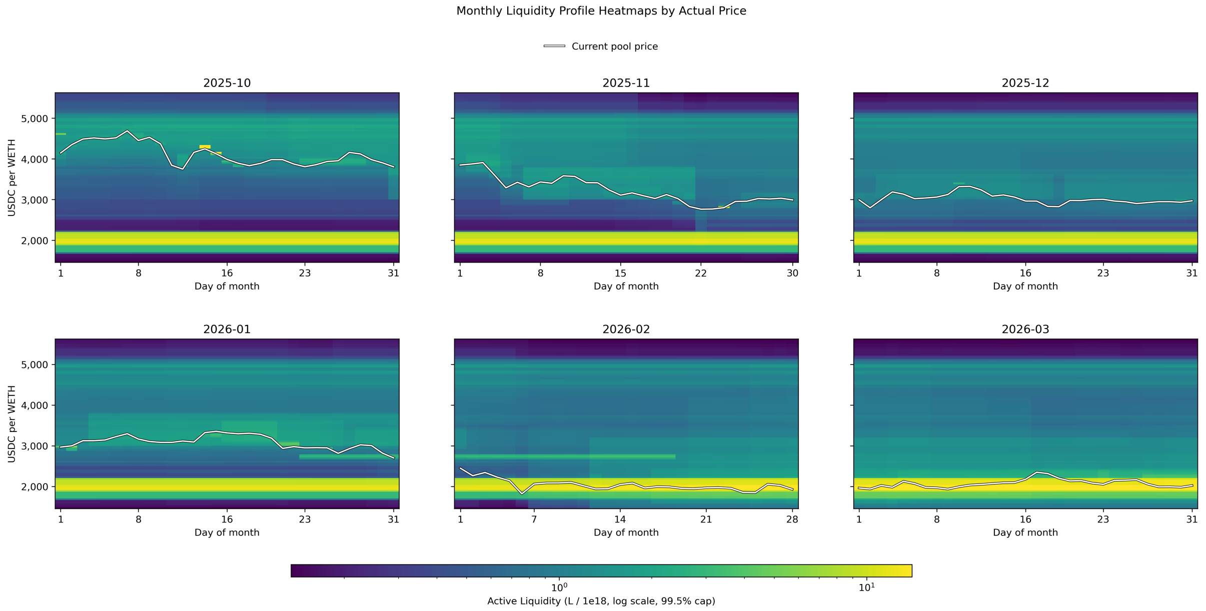

Figure 2: Fig 2.1b — Evolution of the liquidity profile across daily snapshots. Color intensity is active liquidity on a log scale; the white line is the current pool price.

3.2 Task 2.2 — TVL Computation and Decomposition

Total Value Locked (TVL) is a standard measure of the amount of capital currently deposited in a DeFi protocol. In the context of this pool, TVL represents the total dollar value of USDC and WETH supplied by liquidity providers. It gives a first indication of the pool's depth: higher TVL generally means more capital is available to support trades, although it is important to note that in Uniswap V3 the location of that liquidity across price ranges is just as important as the total amount.

We compute TVL from the active LP ranges implied by all mint and burn events up to each daily snapshot. For a position with liquidity L, lower and upper raw square-root prices $s_a < s_b$, and current raw square-root price s, the Uniswap V3 virtual reserve formulas are

$$(x, y) = \begin{cases} \left(L\dfrac{s_b - s_a}{s_a s_b},\, 0\right), & s \le s_a, \\ \left(L\dfrac{s_b - s}{s s_b},\, L(s - s_a)\right), & s_a < s < s_b, \\ (0,\, L(s_b - s_a)), & s \ge s_b, \end{cases}$$

where x is token0 (USDC raw units) and y is token1 (WETH raw units). We convert raw units into decimal-adjusted token amounts and value each position as

$$V = \frac{x}{10^6} + P_{\mathrm{USDC/WETH}} \frac{y}{10^{18}}.$$

Each position is then classified as in range, above the current USDC/WETH price range, or below it.

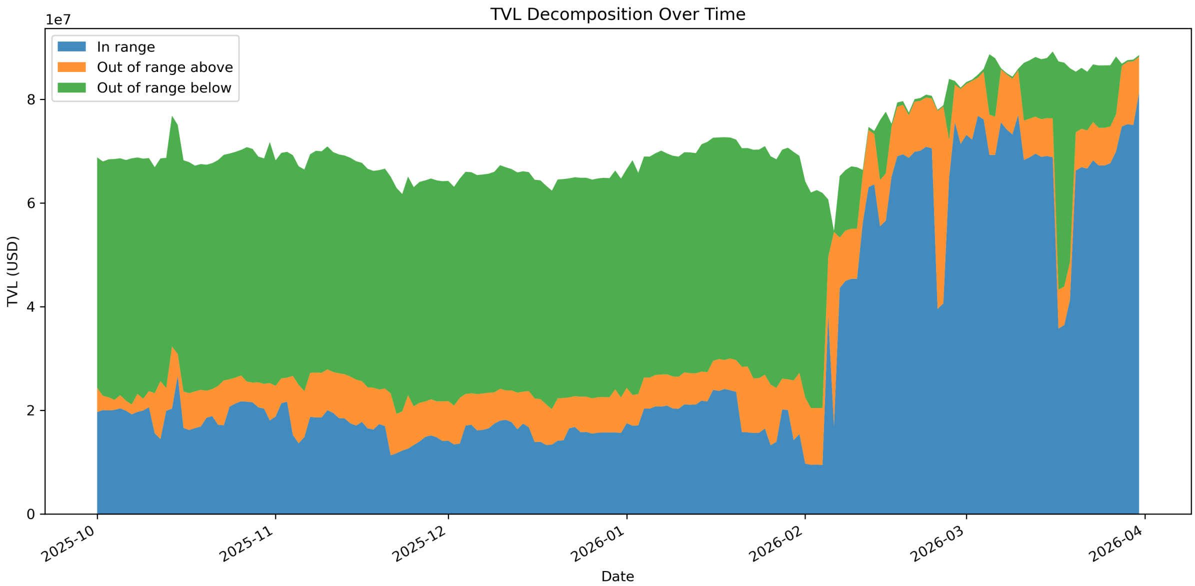

Figure 3: Fig 2.2 — TVL decomposition into in-range and out-of-range liquidity across all daily snapshots.

Average TVL over the sample is \$71.8 million, with a minimum of \$54.6 million on 6 February 2026 and a maximum of \$89.2 million on the 16th of March 2026. The composition changes sharply over time. On the 1st of October 2025, only 28.5% of TVL was in range, while 64.6% was below range. On the 6th of February 2026, in-range TVL was still only 30.9%, with 68.8% above range. By the 31st of March 2026, however, 91.8% of TVL was in range. This is due to 2 main reasons: the price decline into February and March meant that the positions around \$2,000 mentioned in the previous section became in range, and some migration that followed the price decline. The TVL composition is consistent with the liquidity profile visualizations in Figure 1 and the heatmaps in Figure 2, which show a clear migration of liquidity towards the current price over time.

3.3 Task 2.3 — Liquidity Concentration Metrics

3.3.1 Metric A — In-Range Liquidity Ratio (ILR)

The ILR(k) is defined as the fraction of total active liquidity located within a symmetric price band of ±k% around the current price. We will show the ILR for k= 0.1%, 0.5%, 1%, 2% and 5%.

We used the formula

$$\mathrm{ILR}(k)_t = \frac{\sum_i L_{i,t}\mathbf{1}\{P_{i,t} \in [P_t(1-k), P_t(1+k)]\}}{\sum_i L_{i,t}},$$

for $k \in \{0.1\%, 0.5\%, 1\%, 2\%, 5\%\}$. Figure 4 plots all five series.

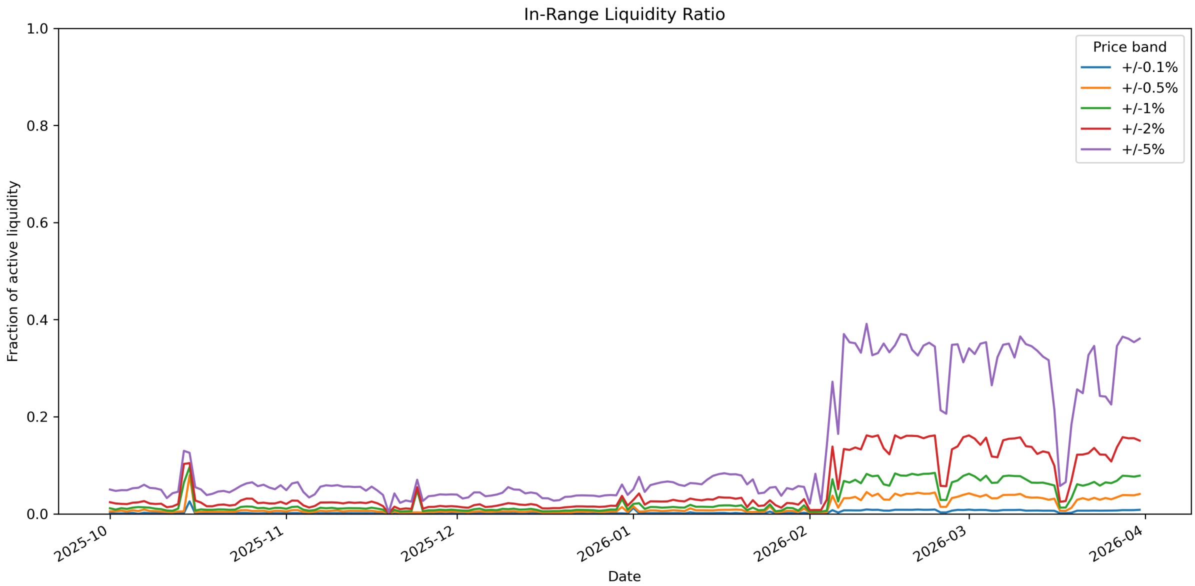

Figure 4: Fig 2.3.A — In-range liquidity ratio for symmetric price bands around the current pool price.

The ILR results show that liquidity near the marginal price increased substantially over the sample. The 5% ILR averaged 12.9%, but rose from 5.0% on the 1st of October 2025 to 36.1% on the 31st of March 2026. The 1% ILR rose from 1.14% to 7.85% over the same dates. This supports the visual evidence that liquidity became more tightly centered around the realized trading range in February and March due to the price declining into the $2000 price band..

3.3.2 Metric B —Liquidity Herfindahl-Hirschman Index (L-HHI)

The L-HHI is defined as the sum of squared liquidity shares across all initialised ticks,

$$\mathrm{HHI} = \sum_i \left(\frac{L_i}{\sum_j L_j} \right)^2 .$$

An L-HHI close to 1 indicates liquidity concentrated in very few ticks, while a value close to 0 indicates near-uniform distribution.

We computed the L-HHI for each daily snapshot using the following formula:

$$\mathrm{LHHI}_t = \sum_i \left(\frac{L_{i,t}^{gross}}{\sum_j L_{j,t}^{gross}} \right)^2 .$$

We overlay the L-HHI with the ETH price in Figure 5.

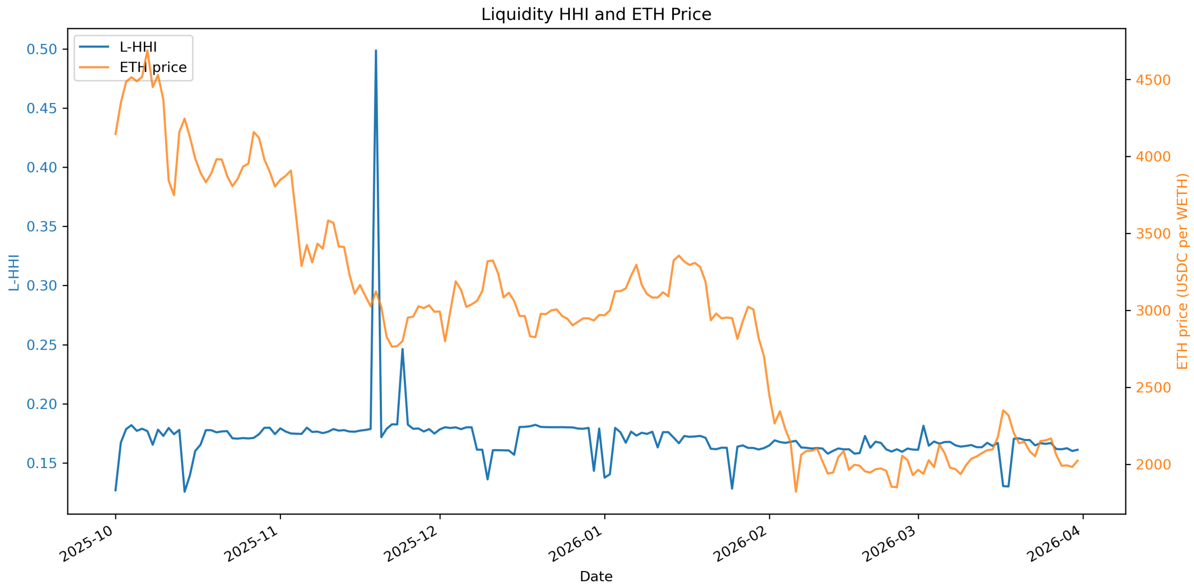

Figure 5: Fig 2.3.B — Liquidity HHI and ETH price.

The average L-HHI is 0.171, with a minimum of 0.126 and a maximum of 0.499. Unlike ILR, L-HHI does not move mechanically with the ETH price: the correlation between price and L-HHI is only 0.12, because the L-HHI measures concentration across initialized ticks globally, whereas ILR measures how much liquidity is close to the current price. Together, the two metrics show that late-sample execution liquidity improved mainly because the current price migrated toward the existing $2000 liquidity band, and some of the LPs moved liquidity with it.

4 Module 3 — Slippage Simulation and Execution Cost

Execution quality in a concentrated-liquidity AMM is shaped by where liquidity sits around the current price. A deep pool can still be expensive to trade against if its liquidity is far away from the active range, while a smaller pool can offer tight execution if capital is concentrated near the marginal price. This module turns the reconstructed Uniswap V3 state from Module 1 into a taker-side execution model for the USDC/WETH 0.05% pool. Our study window runs from 1st October 2025 to 31st March 2026.

4.1 Data Inputs

Our simulator uses two state files from Module 1, with the liquidity snapshot file providing the initialized ticks, `liquidityNet`, and active liquidity at each daily snapshot. The slot0 snapshot file provides the matching `sqrtPriceX96`, current tick, and human-readable mid price. There are 182 daily snapshots in our sample.

For the observed-swap comparison we use `swap_events.parquet`, which contains 1,095,286 swaps. Effective spreads require the pool mid price just before the swap. Since all swaps in a given block share the same previous-block pool state, the natural lookup unit is the unique swap block, not the individual swap. The full file contains 637,982 unique swap blocks. To make the archive-node extraction tractable, we use a documented stratified sample of 25,000 unique swap blocks for the empirical comparison. The sampling metadata is saved in `data/results/module_3/swap_mid_prices.metadata.json`.

The subset was chosen at the block level and stratified by date, trade direction, and USD notional bucket. This preserves the three dimensions that matter most for the comparison:

time variation in liquidity, buy-sell asymmetry, and trade-size composition. It is also aligned with the estimator: because p^{mid} is defined at the previous block, including more swaps from the same block would add event observations but not new mid-price information. The resulting sample is therefore appropriate for comparing median effective spreads by size bucket, while avoiding a much larger archive RPC job.

All prices are reported as USDC per WETH. Internally, token0 is USDC with six decimals and token1 is WETH with eighteen decimals. A `buy_weth` trade means USDC enters the pool and WETH leaves the pool; a `sell_weth` trade means WETH enters and USDC leaves.

4.2 Task 3.1 — Uniswap V3 Swap Simulator

4.2.1 Algorithm

The simulator follows the native Uniswap V3 execution logic. A swap is broken into local steps, each inside a tick interval where active liquidity is constant. The implementation lives in `src/module3/swap_simulator.py`. The core function, `simulate_swap`, is deliberately in-memory: it receives one reconstructed liquidity map, the matching slot0 state, a direction, a USD notional, and the fee. Parquet loading is handled only by a thin convenience wrapper, so the swap logic can be tested without depending on file I/O.

For each trade, the simulator initializes the state from the current `sqrtPriceX96`, current tick, active liquidity, and sorted initialized ticks. It then repeats the following loop until the requested input notional has been consumed:

1. find the next initialized tick in the trade direction;
2. compute the net input needed to move exactly to that tick boundary;
3. deduct the 0.05% pool fee from gross input before computing output;
4. either finish the trade inside the interval or cross the boundary;
5. if a boundary is crossed, update active liquidity using `liquidityNet`.

The active-liquidity update follows the Uniswap V3 tick convention. When WETH is sold into the pool, the internal square-root price moves upward and the liquidity update is $L \leftarrow L + \text{liquidityNet}$. When WETH is bought from the pool, the internal square-root price moves downward and the update is $L \leftarrow L - \text{liquidityNet}$. This sign convention can look counterintuitive in USDC/WETH terms, because the Uniswap raw price is in token1/token0 units and our plotted price is the decimal-adjusted reciprocal.

4.2.2 Formulas

Let $s = \sqrt{P}$ denote the raw Uniswap square-root price and let L denote active liquidity in the current interval. The within-interval token amounts come from the standard Uniswap V3 virtual reserve formulas[1]:

$$\Delta x = L\left(\frac{1}{s_a} - \frac{1}{s_b}\right), \qquad \Delta y = L(s_b - s_a),$$

where x is token0 (USDC), y is token1 (WETH), and $s_a < s_b$ are the square-root prices spanning the step.

For a buy of WETH, token0 is the input and the internal square-root price falls from s to s':

$$\Delta x_{\text{net}} = L\left(\frac{1}{s'} - \frac{1}{s}\right), \qquad \Delta y_{\text{out}} = L(s - s'), \qquad \frac{1}{s'} = \frac{1}{s} + \frac{\Delta x_{\text{net}}}{L}.$$

[1] Uniswap V3 whitepaper, `https://uniswap.org/whitepaper-v3.pdf`.

Table 4: Fig 3.1 — Simulator validation on observed swaps occurring one block after a daily snapshot.

Date	Direction	Notional USD	Observed price	Simulated price	Output error %
2025-10-01	buy WETH	2,013	4,147.71	4,147.71	3.33×10^{-8}
2025-10-06	sell WETH	507	4,515.18	4,515.18	1.61×10^{-7}
2025-10-09	buy WETH	1,250	4,531.61	4,531.61	7.09×10^{-8}
2025-10-09	sell WETH	30,141	4,526.23	4,526.16	-1.62×10^{-3}
2025-10-11	sell WETH	47,027	3,837.62	3,837.62	3.50×10^{-9}
2025-10-22	sell WETH	24,482	3,870.43	3,870.43	1.50×10^{-9}

For a sale of WETH, token1 is the input and the internal square-root price rises:

$$\Delta y_{\text{net}} = L(s' - s), \qquad \Delta x_{\text{out}} = L\left(\frac{1}{s} - \frac{1}{s'}\right), \qquad s' = s + \frac{\Delta y_{\text{net}}}{L}.$$

The fee is charged on input at every step:

$$\Delta q_{\text{net}} = (1 - f)\Delta q_{\text{gross}}, \qquad f = 0.0005.$$

Average execution prices include this fee. We report price impact including fees and slippage net of the 5 bps fee:

$$\text{slippage}_{bps} = \text{price impact}_{bps} - 5.$$

For buys, $p^{exec} = \text{USDC}_{in}/\text{WETH}_{out}$ and

$$\text{impact}_{bps} = \left(\frac{p^{exec}}{p^{mid}} - 1\right) 10{,}000.$$

For sells, $p^{exec} = \text{USDC}_{out}/\text{WETH}_{in}$ and

$$\text{impact}_{bps} = \left(1 - \frac{p^{exec}}{p^{mid}}\right) 10{,}000.$$

This gives positive execution costs for both directions.

4.2.3 Validation Against Observed Swaps

To validate the simulator against real pool behavior, we use observed swaps that occur exactly one block after a daily snapshot block. For these swaps, the reconstructed daily snapshot is the pre-swap state, so the simulator can be run with the actual observed input and compared directly with the event output. This gives us 112 validation swaps across 81 distinct days. The median absolute output error is $3.33 \times 10^{-8}\%$, the 95th percentile is 0.123%, and a maximum of 0.188%. Notably, the small non-zero tail is concentrated in blocks with multiple swaps, where later swaps in the block are affected by earlier swaps from the same block.

4.3 Task 3.2 — Simulation Grid

We evaluate the simulator over a grid that spans small retail-sized trades and large institutional-sized trades:

$$\{1\text{K}, 10\text{K}, 50\text{K}, 100\text{K}, 250\text{K}, 500\text{K}, 1\text{M}\} \times \{\text{buy WETH}, \text{sell WETH}\} \times 182 \text{ snapshots}.$$

This produces 2,548 simulated trades, saved in `simulated_trades.parquet`. Each row records the snapshot block and timestamp, direction, notional USD, average execution price, pool mid

Table 5: Median simulated price impact by trade size.

Notional USD	Buy WETH median bps	Sell WETH median bps
1,000	5.10	5.10
10,000	5.97	5.97
50,000	9.84	9.87
100,000	14.67	14.69
250,000	29.10	29.29
500,000	53.49	53.43
1,000,000	102.04	102.56

price, price impact in bps, slippage in bps net of the 5 bps fee, and the number of tick boundaries crossed. We also store token input/output amounts and the final simulated price/tick as diagnostics.

The generated grid passes a set of consistency checks. Every snapshot has exactly 14 simulated trades, all sizes and directions are present, required metrics are non-missing, price impact is positive, and slippage equals price impact minus 5 bps up to numerical precision. Price impact is also monotone in trade size within every snapshot-direction pair. Median tick-boundary crossings rise from zero at $1K and $10K trades to 18 at $1M trades, which is exactly the pattern expected when larger trades consume liquidity across more initialized ranges.

4.4 Task 3.3 — Price Impact Curves

The first empirical question is how quickly execution cost grows with trade size. From the simulated trade grid, we compute the median, 10th percentile, and 90th percentile of price impact by direction and notional size. Figure 6 shows these curves on log-log axes.

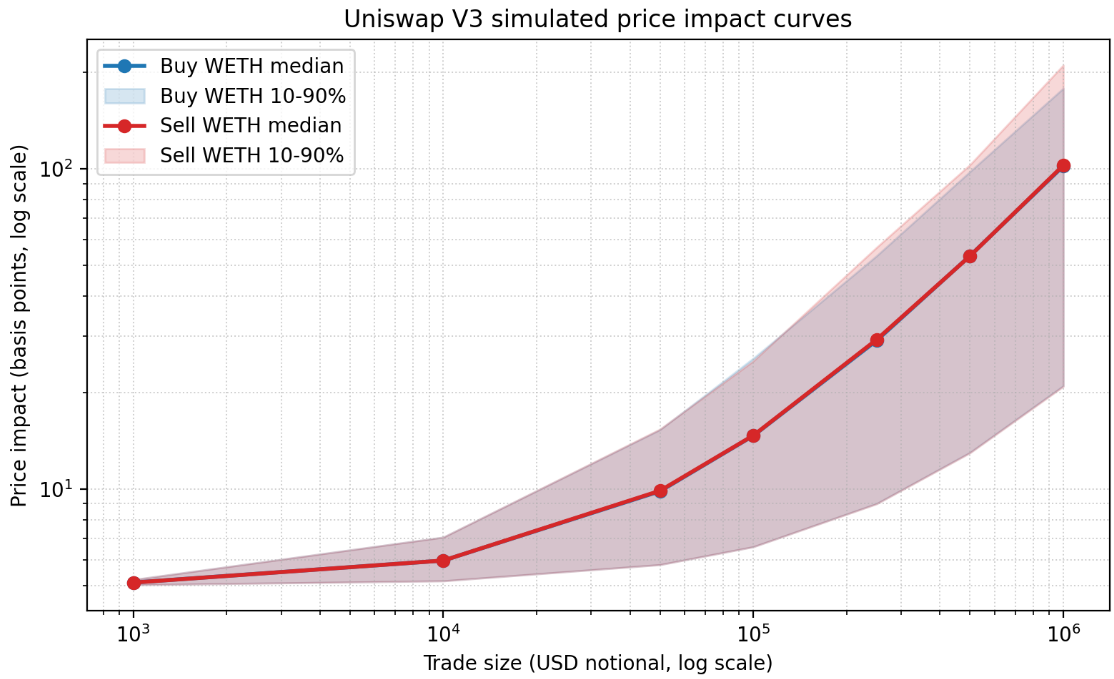

Figure 6: Fig 3.2 — Simulated price impact by trade size. Solid lines show the median across daily snapshots; shaded areas show the 10th–90th percentile range.

The buy and sell curves are almost identical in the median snapshot, suggesting that liquidity immediately around the current price is broadly balanced on the two sides of the book. The cost curve is strongly convex. Moving from \$10K to \$100K raises median impact from about 6 bps to about 15 bps, while moving from \$100K to \$1M raises it to roughly 102 bps. Dispersion also grows with trade size: for \$1M sells, the 10th–90th percentile band spans about 21 to 211 bps. At large sizes, execution quality is therefore highly state dependent; the same notional can be cheap or expensive depending on how tightly liquidity is distributed around the daily price.

4.5 Task 3.4 — Effective Spread from Observed Swaps

The simulations give counterfactual execution costs against daily pool states. As a complement, we estimate realised effective spreads directly from observed swaps. For swap i,

$$ES_i = 2D_i \frac{p_i^{exec} - p_i^{mid}}{p_i^{mid}},$$

where $D_i = +1$ for a buy and $D_i = -1$ for a sell. The execution price is implied by the swap amounts:

$$p_i^{exec} = \frac{|\text{amount0\_usdc}_i|}{|\text{amount1\_weth}_i|}.$$

The mid price p_i^{mid} is the pool price from `slot0()` at the block before the swap. Swaps are then assigned to the same seven notional buckets as the simulation grid, using geometric midpoints between adjacent bucket sizes as cutoffs.

The use of a stratified subset matters most in this section. A full pass would require one historical `slot0()` call for each unique swap block, which is a large archive-node job. Instead, we sample 25,000 unique blocks while preserving coverage across dates, trade directions, and notional buckets. This design is well suited to the reported statistic, because Fig. 7 compares medians by bucket rather than individual swap paths. Stratification prevents the sample from being dominated by the most common small trades or by high-activity days, and block-level sampling avoids overweighting blocks that contain many swaps but only one previous-block mid price.

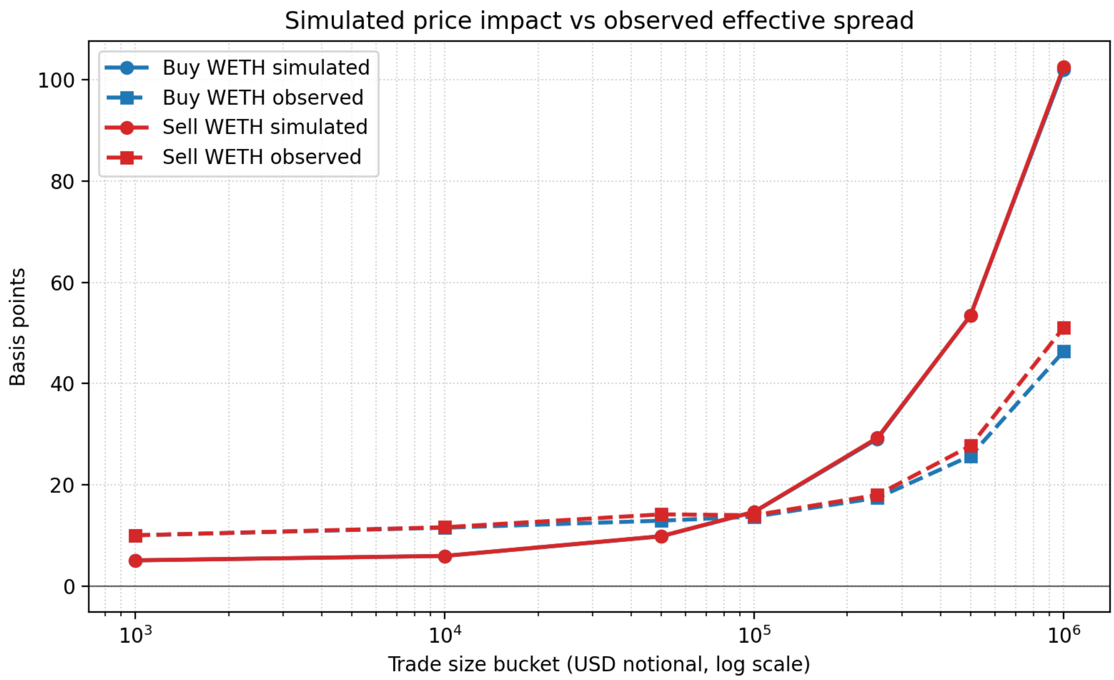

Figure 7: Fig 3.3 — Median observed effective spread by trade-size bucket compared with the simulated median price impact.

The observed effective-spread curves increase with trade size, matching the direction of the simulator. For small and medium trades, observed spreads sit above simulated impacts. At \$1K–\$10K, the simulator is close to the 5 bps fee floor, while observed spreads are around 10–12 bps. This gap is consistent with effects that the static simulator intentionally leaves out: within-block ordering, short-term price movement, routing, and MEV.

At the largest bucket, the comparison reverses. The simulation predicts about 102 bps for an isolated \$1M trade against the daily pool state, while the sampled observed spreads are closer to 45–50 bps. A plausible interpretation is that very large realised trades are not random market orders submitted into one static pool state. They may be split, routed, timed, or executed in blocks where liquidity is temporarily deeper than the daily median snapshot. In that sense, the simulator measures mechanical single-pool depth, while the effective spread captures realised execution after traders and routers have optimized around that depth.

4.6 Limitations

The simulator isolates the mechanical cost of trading through the Uniswap V3 pool. It does not model gas, latency, private order flow, multi-hop routing, same-block ordering, or strategic order splitting. The simulation grid also uses one daily state per day, whereas observed swaps arrive continuously. The effective-spread comparison is therefore best read as an empirical benchmark for the simulator rather than a one-for-one match for every swap.

The pre-swap mid-price extraction for Fig. 7 uses a stratified sample rather than all 637,982 unique swap blocks. This is a practical compromise, but not an arbitrary one: the sample keeps the dimensions needed for the analysis and is large enough to estimate medians in the reported size-direction buckets. A full extraction would reduce sampling uncertainty, but it would not change the mechanics of the estimator.

4.7 Takeaway

The main result is that concentrated liquidity provides very tight execution for small trades and increasingly expensive execution for large trades. Around the median daily state, $1K trades cost about 5.1 bps including the pool fee, while $1M trades cost roughly 102 bps. Observed effective spreads follow the same upward pattern, but they also reveal the gap between a static pool simulation and realised execution in the mempool and routing environment. The simulator is therefore useful as a clean measure of pool depth, while the effective-spread exercise shows how actual traders interact with that depth.

5 Module 4 — Liquidity Provision Analytics

5.1 Objective and Scope

This module studies the USDC/WETH Uniswap V3 pool from the liquidity provider (maker) perspective. We construct five synthetic representative LP positions sized at $100,000 at the start of the study window, then decompose performance into (i) fee income earned while the position is in-range and (ii) impermanent loss (IL) relative to a HODL benchmark holding the same initial token quantities.

Pool configuration used in this module.

- Protocol: Uniswap V3
- Pair: USDC/WETH
- Fee tier: 0.05% (5 bps)
- Contract: `0x88e6A0c2dDD26FEEb64F039a2c41296FcB3f5640`

5.2 Data Inputs

Inputs used.

- `slot0_snapshots.parquet`: daily snapshot blocks with mid-price (USDC per WETH) and current tick.
- `liquidity_snapshots.parquet`: tick-level liquidity map at the first snapshot, used to initialize fee replay.
- `mint_burn_events.parquet`: historical liquidity updates, replayed in log order to keep the liquidity map current.
- `swap_events.parquet`: per-swap token deltas and post-swap state, used to replay fee accrual interval by interval.

5.3 Task 4.1 — Representative LP Positions

We define five synthetic positions, all entered at the first snapshot and exited at the last snapshot. Positions P1–P4 are symmetric price bands around the entry price, while P5 is full range (V2-equivalent).

Position set (constructed at entry).

Table 6: Representative LP positions at entry (notional \$100,000).

ID	Tick lower	Tick upper	Initial USDC	Initial WETH
P1	193010	193040	60,254.76	9.5874
P2	192970	193080	52,793.93	11.3871
P3	192820	193230	50,746.78	11.8809
P4	192060	194080	52,320.62	11.5013
P5	-887270	887270	50,000.00	12.0611

Liquidity sizing under the \$100,000 constraint. To avoid ambiguity with the project brief's token notation, we use C for USDC and E for WETH throughout this module. Let the entry price be p_0 (USDC per WETH), with corresponding square-root price $\sqrt{p_0}$. For a range with bounds $(\sqrt{p_a}, \sqrt{p_b})$, Uniswap V3 virtual reserves for liquidity L imply:

$$C(L) = \begin{cases} L\left(\frac{1}{\sqrt{p_a}} - \frac{1}{\sqrt{p_b}}\right), & \sqrt{p_0} \le \sqrt{p_a}, \\ L\left(\frac{1}{\sqrt{p_0}} - \frac{1}{\sqrt{p_b}}\right), & \sqrt{p_a} < \sqrt{p_0} < \sqrt{p_b}, \\ 0, & \sqrt{p_0} \ge \sqrt{p_b}, \end{cases}$$

and

$$E(L) = \begin{cases} 0, & \sqrt{p_0} \le \sqrt{p_a}, \\ L(\sqrt{p_0} - \sqrt{p_a}), & \sqrt{p_a} < \sqrt{p_0} < \sqrt{p_b}, \\ L(\sqrt{p_b} - \sqrt{p_a}), & \sqrt{p_0} \ge \sqrt{p_b}, \end{cases}$$

where C is the amount of token0 (USDC) and E is the amount of token1 (WETH). The entry portfolio value is $V_0(L) = C(L) + p_0\, E(L)$. We solve for L via:

$$L = \frac{100{,}000}{V_0(1)},$$

and then compute initial deposits $(C_0, E_0) = (C(L), E(L))$.

5.4 Task 4.2 — Fee Income Computation

The project brief describes the Uniswap V3 fee-accumulator mechanism. Since our LPs are synthetic and therefore have no historical `feeGrowthInside` state, we compute an interval-level counterfactual fee estimate that is equivalent to allocating each swap interval's fees by active liquidity share. Starting from the first daily liquidity snapshot, we replay historical events in log order, update tick liquidity with every Mint/Burn event, and split each Swap across both initialized pool ticks and synthetic LP range boundaries. This avoids treating a cross-range swap as if it occurred at only its final tick. The historical swap path is kept fixed, so this is a counterfactual fee allocation rather than a full re-simulation of market execution with the synthetic LP added.

Relationship to the brief's prescribed formula. The project brief prescribes allocating each swap's fees in proportion to L_j/L_{active}, where L_{active} is the active liquidity recorded in the Swap event's `liquidity` field. We report this brief-style calculation as a baseline, but implement it at interval level rather than at the final Swap-event tick. The reason is that the Swap event's `liquidity` field is the *post-swap* active liquidity; for any swap that crosses one or more initialized ticks it does not describe the liquidity in force over the early part of the trade. Our interval-level replay therefore charges each sub-interval at the historical pool liquidity actually active there, while still preserving the brief's fee-growth denominator.

We then add a second, more conservative counterfactual calculation. Our LPs are synthetic and were never part of the historical pool, so a synthetic entrant of size L_j would itself have diluted every other LP's share. In this adjusted version we place L_j in the fee-growth denominator $(L_{\text{pool}} + L_j)$. Each representative position is evaluated independently, so the adjusted denominator includes the historical pool liquidity plus that one synthetic LP, not all five synthetic LPs at once.

Interval fee growth. With fee rate $f = 0.0005$, each swap interval contributes fees in exactly one input token. For interval k of swap i, let $\Delta C^{+}_{i,k}$ be the USDC input amount and $\Delta E^{+}_{i,k}$ the WETH input amount, both in token units after splitting the swap path by tick interval. The token fees paid to the pool are:

$$F^{C}_{i,k} = f\,\Delta C^{+}_{i,k}, \qquad F^{E}_{i,k} = f\,\Delta E^{+}_{i,k}.$$

The implementation then applies the fee-accumulator logic at interval level: fees are first expressed as token fee growth per unit active liquidity, and the synthetic LP's token fees are obtained by multiplying that growth by its liquidity.

Synthetic LP fee growth. For position j with liquidity L_j, interval k earns fees only if its tick is in $[\text{tick}^{(j)}_{\text{lower}}, \text{tick}^{(j)}_{\text{upper}})$. The brief-style fee-growth calculation is:

$$\Delta g^{C,\text{brief}}_{i,k,j} = \frac{F^{C}_{i,k}}{L_{\text{pool},k}}, \qquad \Delta g^{E,\text{brief}}_{i,k,j} = \frac{F^{E}_{i,k}}{L_{\text{pool},k}}.$$

The synthetic-liquidity-adjusted calculation is:

$$\Delta g^{C,\text{adj}}_{i,k,j} = \frac{F^{C}_{i,k}}{L_{\text{pool},k} + L_j}, \qquad \Delta g^{E,\text{adj}}_{i,k,j} = \frac{F^{E}_{i,k}}{L_{\text{pool},k} + L_j}.$$

For either variant $v \in \{\text{brief}, \text{adj}\}$, token fees are:

$$\text{Fee}^{C,v}_{i,k,j} = L_j\,\Delta g^{C,v}_{i,k,j}, \qquad \text{Fee}^{E,v}_{i,k,j} = L_j\,\Delta g^{E,v}_{i,k,j}.$$

We aggregate token fees between daily snapshot blocks and report cumulative USDC fees, cumulative WETH fees, and their USD value for both variants:

$$\text{FeeUSD}^{v}_{j}(t) = \sum_{i,k \le t} \left(\text{Fee}^{C,v}_{i,k,j} + p_{i,k}\,\text{Fee}^{E,v}_{i,k,j} \right),$$

where $p_{i,k}$ is the USDC/WETH price during the swap interval. This matches the code path that converts interval WETH fees to USD at the interval price before cumulating the USD fee series. The brief converts token fees "at the ETH price prevailing at the time of collection"; because a synthetic LP never issues a `Collect`, we instead price each WETH fee at the moment it accrues (the swap-interval price), which is the natural choice for continuous counterfactual accrual and coincides with collection-time pricing in the limit of frequent collection.

5.5 Task 4.3 — Impermanent Loss and Net P&L

At each daily snapshot t with ETH price p_t, we compute:

$$V_{\text{HODL}}(t) = C_0 + p_t\,E_0, \qquad V_{\text{LP}}(t) = C_t + p_t\,E_t, \qquad \text{IL}(t) = V_{\text{HODL}}(t) - V_{\text{LP}}(t),$$

where (C_t, E_t) are the Uniswap V3 virtual reserves implied by the fixed liquidity L and the current price relative to the position range (below, within, or above). Finally, we report a net metric consistent with the project brief for both fee variants:

$$\text{Net}^{v}(t) = \text{FeeUSD}^{v}(t) - \text{IL}(t), \qquad v \in \{\text{brief}, \text{adj}\}.$$

5.6 Results

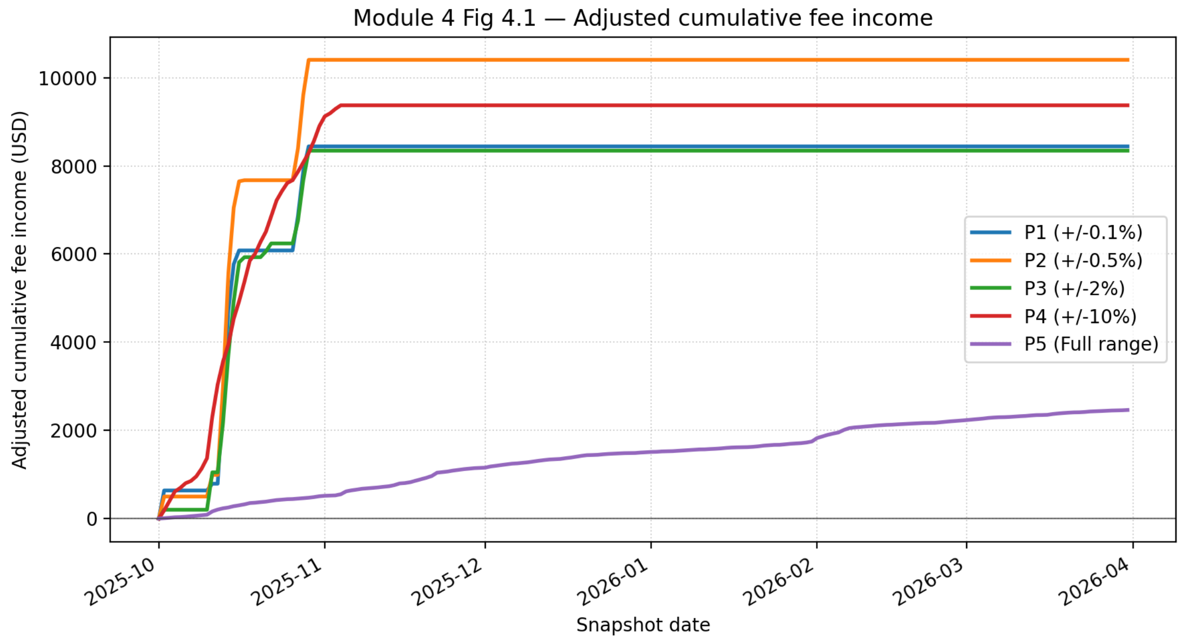

Figure 8: Fig 4.1 — Adjusted cumulative fee income time series for the five representative LP positions.

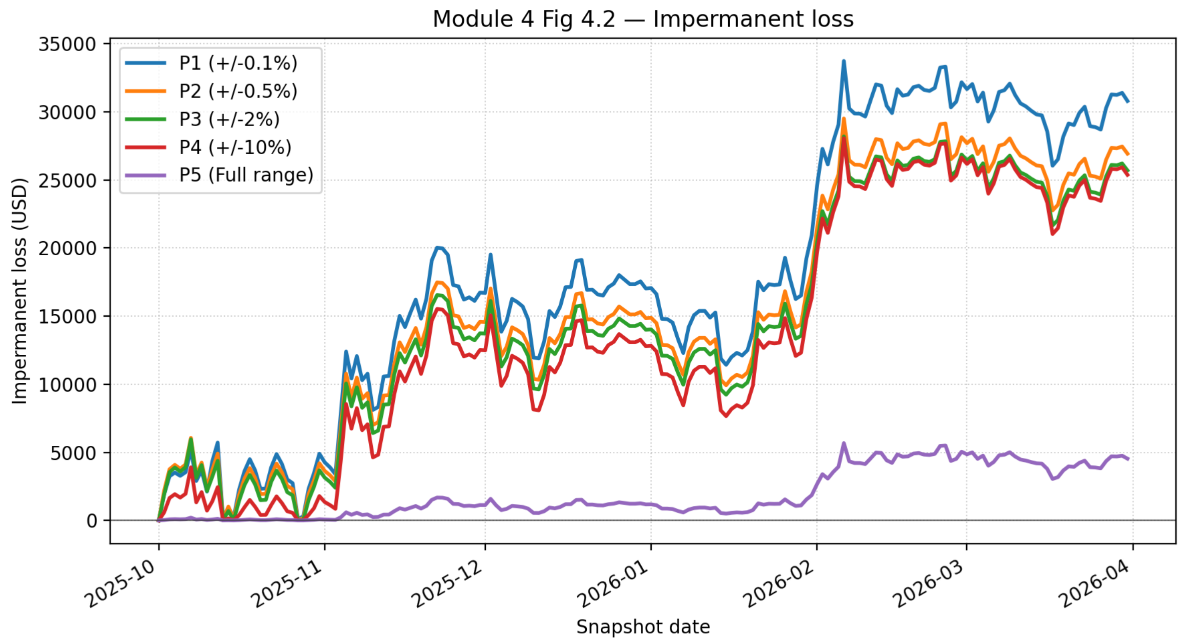

Figure 9: Fig 4.2 — Impermanent loss time series (HODL value minus LP principal value).

Table 7: Module 4 final LP performance by representative position.

ID	In-range	Brief fees	Adj. fees	IL	Brief net	Adj. net
P1	1	13,885.44	8,441.29	30,800.18	-16,914.74	-22,358.88
P2	4	12,276.66	10,403.54	26,934.79	-14,658.13	-16,531.25
P3	5	8,775.01	8,343.72	25,703.24	-16,928.24	-17,359.53
P4	33	9,484.54	9,372.37	25,379.44	-15,894.90	-16,007.07
P5	182	2,464.76	2,463.26	4,534.54	-2,069.78	-2,071.28

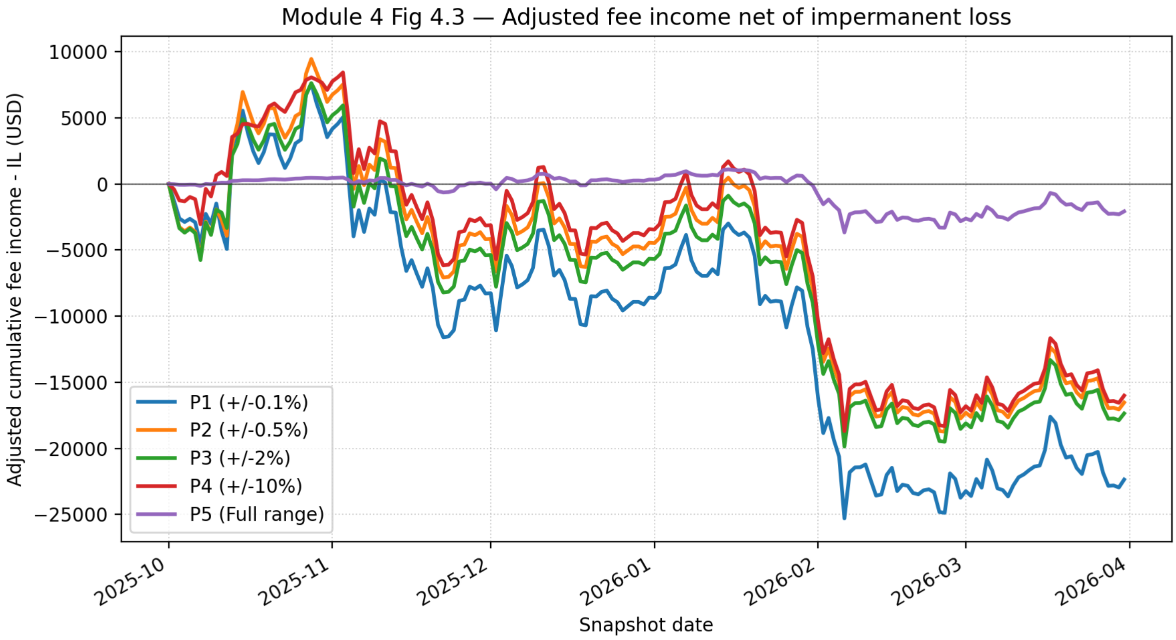

Figure 10: Fig 4.3 — Adjusted cumulative fee income minus impermanent loss for each position.

Final performance summary. Table 7 reports the end-of-window values on 31 March 2026. "Brief fees" use the project brief's historical pool denominator, while "Adj. fees" add the representative LP's synthetic liquidity to the denominator. The pool price fell from \$4,145.56 per WETH at entry to \$2,024.66 at exit, so all concentrated LP positions ended above their upper WETH price range and were fully converted into WETH. This explains why fee income is not enough to offset IL for P1–P4: the positions earn fees only while active, but the final mark-to-market loss relative to HODL remains large after the price move.

This fixed-range construction should therefore be read as a passive benchmark, not as a professional active LP strategy. A fixed Uniswap V3 range is closer to a finite inventory range order than to market making forever: once the price leaves the range, the LP earns no further fees, holds one-sided inventory, and continues carrying directional exposure. The experiment isolates the payoff and IL mechanics of concentrated liquidity, but an actively managed LP would periodically recenter, widen, or withdraw liquidity after the market moves away from the position.

The "In-range" column counts the daily midnight snapshots whose pool tick fell inside $[\text{tick}_{\text{lower}}, \text{tick}_{\text{upper}})$. It is only a daily sampling of position state, not the fee-earning window. Fees accrue continuously on the replayed intra-day swap path, so a narrow position such as P1 can earn substantial fees even though the price sat inside its band at only one midnight snapshot, because price crossed back and forth through the band intraday.

The fee ranking illustrates the central trade-off in concentrated liquidity. Under the brief-style denominator, P1 has the highest fee estimate (\$13.9k), because its liquidity is very dense when active. Under the adjusted counterfactual, P1 falls to \$8.4k because the synthetic liquidity is large relative to the historical pool liquidity in that narrow band, so adding it to the denominator materially dilutes its own fee share. P2 earns the most adjusted fees, about \$10.4k, because it still concentrates capital tightly while capturing more flow near the entry price. P3 and P4 spread capital over wider intervals; P4 is active for 33 days and therefore slightly outearns P3 in the adjusted series, but both give up fee share per dollar relative to the narrower bands. P5 is active for the entire period but earns the least, because the full-range position is closest to passive V2-style liquidity and has the lowest fee share per dollar deposited.

The IL profile moves in the opposite direction. The narrow positions are shorter gamma around the initial price: when ETH sells off, they quickly become one-sided WETH positions and underperform the initial HODL basket. Wider ranges soften this effect, but P2–P4 still finish with IL around \$25–27k. The full-range P5 has much smaller IL, about \$4.5k, because it adjusts more gradually and never leaves the active range. Net of fees, every representative position is negative over this window, but P5 loses the least. This is a useful bridge to Module 5: the LP payoff behaves like a short volatility exposure, and fee income is the compensation for bearing that rebalancing risk.

Limitations. The fee replay uses 80-digit decimal arithmetic and does not reproduce Uniswap V3's exact on-chain integer rounding (uint160 square-root prices and per-step truncation). At the notionals and tick spacings in this pool the resulting error is negligible relative to the fee magnitudes reported, but the figures should be read as a high-precision counterfactual estimate rather than bit-exact on-chain accounting. As noted above, the synthetic LP enters the fee-growth denominator but does not alter the historical swap path, so the numbers answer "what would this position have earned and lost," not "how would the market have executed differently had this LP been present."

6 Module 5: Dynamic Hedging of Impermanent Loss

6.1 LP Payoff, Delta, and Gamma

6.1.1 LP Payoff Shape (Task 5.1.1)

Let $s(p) = \sqrt{10^{12}/p}$ denote the raw Uniswap sqrt-price for ETH price p (USDC per WETH), and let $s_a < s_b$ be the position's lower and upper sqrt-price boundaries. For a position with liquidity L, the fee-less LP value is

$$V_{LP}(p) = \begin{cases} \dfrac{L(s_b - s_a)}{s_a\, s_b \cdot 10^6}, & s(p) \le s_a \quad \text{(above range, all USDC)}, \\ \dfrac{L(s_b - s(p))}{s(p)\, s_b \cdot 10^6} + p\, \dfrac{L(s(p) - s_a)}{10^{18}}, & s_a < s(p) < s_b \quad \text{(in range)}, \\ p\, \dfrac{L(s_b - s_a)}{10^{18}}, & s(p) \ge s_b \quad \text{(below range, all WETH)}. \end{cases}$$

Figure 11 plots $V_{LP}(p_T)$ for all five positions over $p_T \in [0.5\, p_0,\ 1.5\, p_0]$, alongside the HODL benchmark $V_{HODL}(p_T) = x_0\, p_T + y_0$. The vertical gap between the two curves is the impermanent loss.

The payoff is concave and resembles a *short straddle*: the LP underperforms the HODL portfolio for any large move in either direction. The analogy is useful but not exact due to the fact that there is no upfront premium. Compensation arrives as swap fees only while the price stays within the active range, and the payoff has finite inventory endpoints rather than the unbounded losses of a written option.

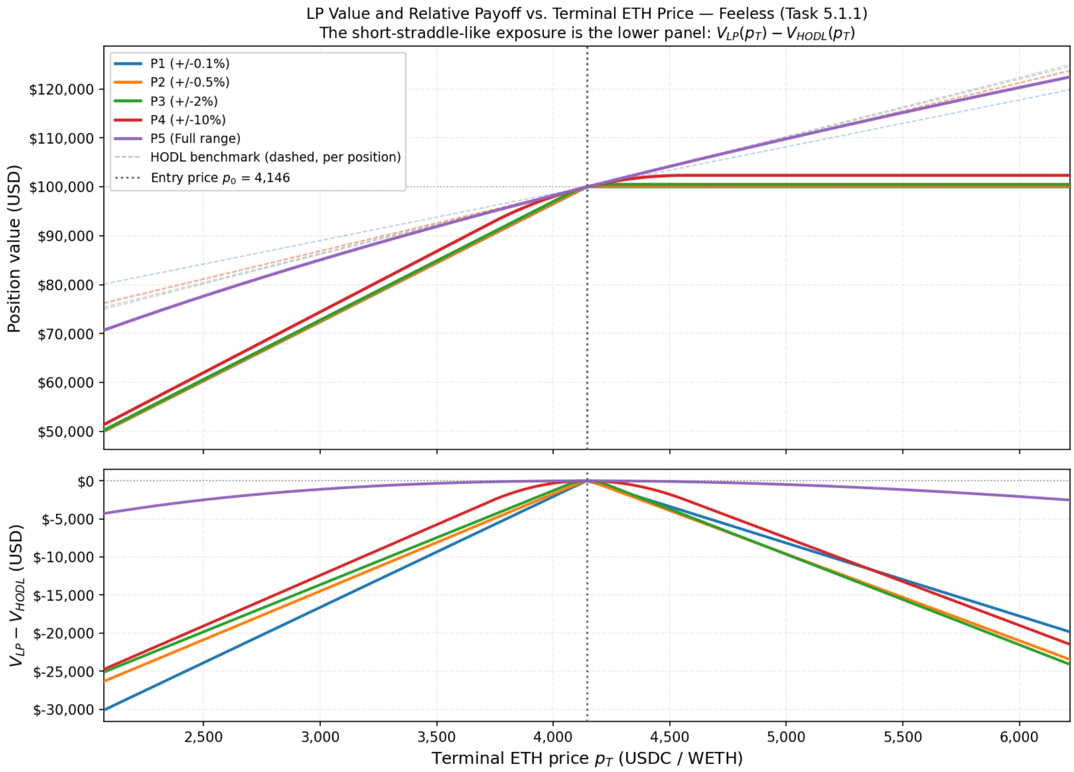

Figure 11: Fee-less LP payoff (solid) vs. HODL benchmark (dashed) for the five representative positions over ±50% of the entry price. The gap is the impermanent loss; narrower positions suffer larger IL for the same price move.

6.1.2 Derivation of LP Delta (Task 5.1.2)

Since $s(p) = (10^{12}/p)^{1/2}$, we have $ds/dp = -s(p)/(2p)$.

Above range ($s(p) \le s_a$): V_{LP} is constant in p, so $\Delta_{LP} = 0$.

Below range ($s(p) \ge s_b$): $V_{LP} = p \cdot L(s_b - s_a)/10^{18}$ is linear in p, so $\Delta_{LP} = L(s_b - s_a)/10^{18}$.

In range: Using $p\,s^2 = 10^{12}$ (equivalently, $1/(2p\,s) = s/(2 \cdot 10^{12})$), differentiate each term of V_{LP} with respect to p:

$$\frac{d}{dp}\left[\frac{L(s_b - s)}{s\,s_b \cdot 10^6}\right] = \frac{L}{10^6} \cdot \frac{d}{dp}\left(\frac{1}{s}\right) = \frac{L}{10^6} \cdot \frac{-1}{s^2} \cdot \frac{ds}{dp} = \frac{L}{10^6} \cdot \frac{-1}{s^2} \cdot \left(-\frac{s}{2p}\right) = \frac{L}{2p\,s \cdot 10^6} = \frac{L\,s(p)}{2 \cdot 10^{18}},$$

$$\frac{d}{dp}\left[\frac{p\,L(s - s_a)}{10^{18}}\right] = \frac{L}{10^{18}}\left[(s - s_a) + p\,\frac{ds}{dp}\right] = \frac{L}{10^{18}}\left[(s - s_a) - \frac{s}{2}\right] = \frac{L(s(p)/2 - s_a)}{10^{18}}.$$

Summing: $\Delta_{LP} = L\,s(p)/(2 \cdot 10^{18}) + L(s(p)/2 - s_a)/10^{18} = L(s(p) - s_a)/10^{18}$.

The complete analytical delta is therefore

$$\Delta_{LP}(p) = \begin{cases} L(s_b - s_a)/10^{18}, & s(p) \ge s_b, \\ L(s(p) - s_a)/10^{18}, & s_a < s(p) < s_b, \\ 0, & s(p) \le s_a. \end{cases}$$

In the active range Δ_{LP} equals the LP's current WETH inventory, which declines continuously as the price rises (WETH is sold into the pool).

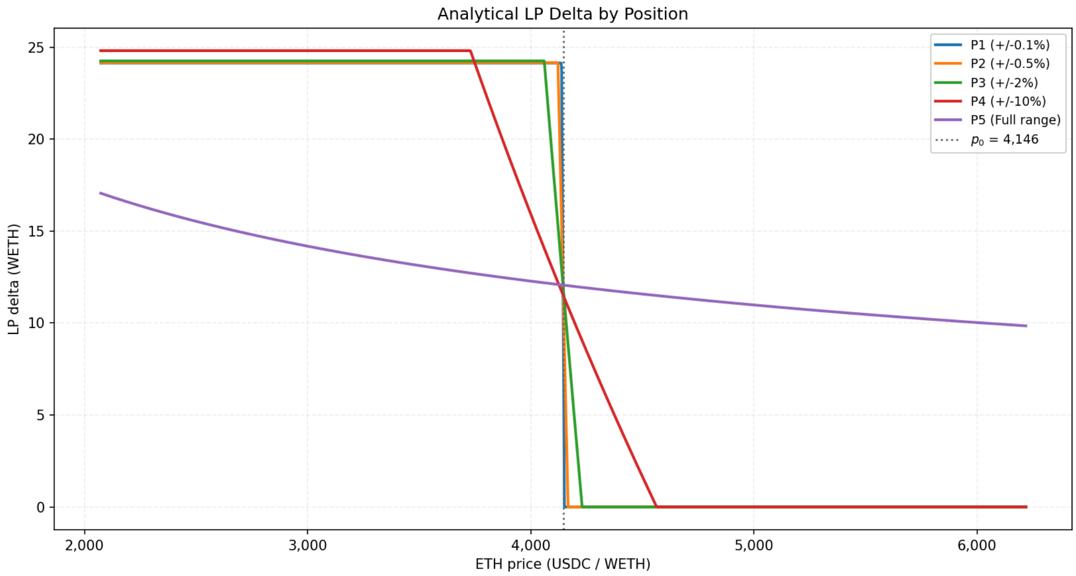

Figure 12: Analytical LP delta by position. Delta is constant below the active range, decreasing linearly through it, and zero above.

6.1.3 LP Gamma (Task 5.1.3)

Differentiating Δ_{LP} once more:

$$\Gamma_{LP}(p) = \begin{cases} \dfrac{-L\, s(p)}{2p \cdot 10^{18}}, & s_a < s(p) < s_b, \\ 0, & \text{otherwise.} \end{cases}$$

Gamma is strictly negative inside the active range: the LP is *short gamma*. Its magnitude $|\Gamma_{LP}| = L\, s(p)/(2p \cdot 10^{18})$ is proportional to L. For a fixed notional entry of \$100 000, a narrower position requires a larger L to cover a smaller price interval, so it carries larger $|\Gamma|$ per dollar invested. The wide positions P4 and P5 have the smallest $|\Gamma|$; P1 (the narrowest, $\pm 0.1\%$) has the largest.

6.2 Hyperliquid Data Collection (Task 5.2)

Both files cover 2025-10-01 to 2026-03-31 with 4 368 hourly rows each.

`perp_prices.parquet`: hourly OHLCV candles for the ETH perpetual on Hyperliquid, retrieved from the 0xArchive REST API. The official Hyperliquid `candleSnapshot` endpoint was not used because its rolling retention cap excluded the first 24 days of the study window.

`funding_rates.parquet`: hourly funding rates from the official Hyperliquid `fundingHistory` endpoint (paginated). Oracle prices, required for the funding P&L formula below, were joined from the 0xArchive hourly price feed. An 18-hour indexer gap on 2026-01-19 was back-filled from Allium's native Hyperliquid context table; no non-native oracle values are used.

The funding payment per hour for a short position holding q ETH is

$$\text{funding P\&L}_t = q \cdot \text{oracle}_t \cdot f_t,$$

where f_t is the hourly funding rate. Positive f_t means the short receives funding; negative f_t means the short pays.

Figure 14 shows the hourly and cumulative funding rate alongside the ETH perp close price over the study window.

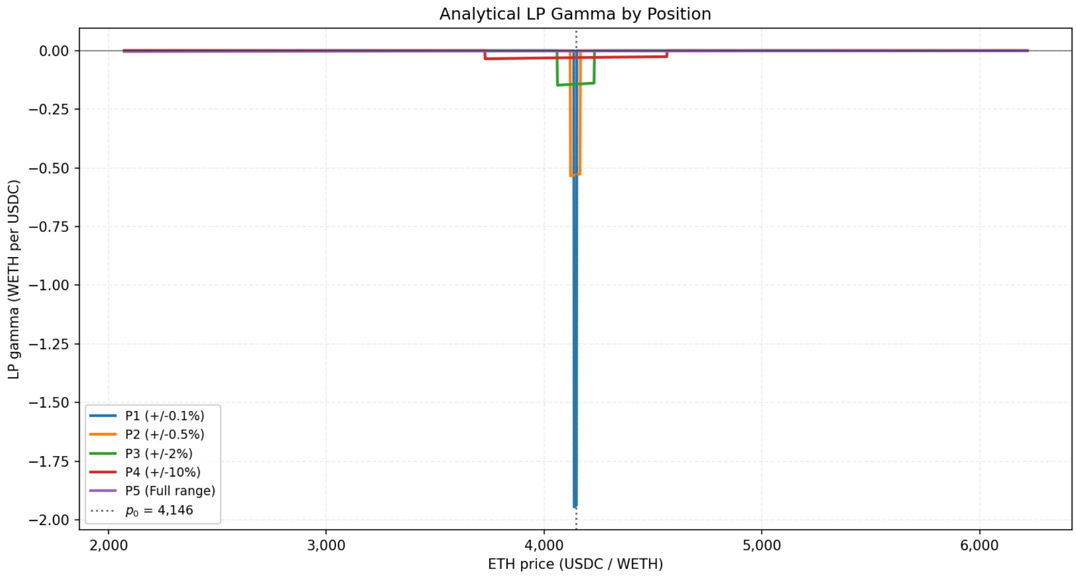

Figure 13: Analytical LP gamma by position. Gamma is zero outside the active range and strictly negative within it; concentrated positions have larger magnitude.

6.3 Delta-Hedging Backtest (Task 5.3)

Strategy. At each rebalance event the strategy opens a short ETH perpetual of size $q_t^{short} = \Delta_{LP}(p_t)$ ETH, offsetting the LP's long WETH exposure. Between rebalances the size is held static. Three rebalancing frequencies are tested: 1h, 4h, and 24h, for each of the five LP positions, giving us $5 \times 3 = 15$ strategy variants.

Monte Carlo expected-delta variant. As an extension, the fixed hedge size $\Delta_{LP}(p_t)$ is replaced by a forward-looking target: $q_t^{short} = \mathbb{E}[\Delta_{LP}(p_{t+h}) \mid p_t]$, where the expectation is estimated by simulating 10,000 GBM paths over the next rebalance horizon h. The rationale is that the current delta will drift before the next rebalance, so targeting the expected future delta pre-hedges some of that gamma exposure. The same P&L accounting and trading-fee structure applies. Results are available in `hedge_results.parquet` under `strategy = monte_carlo_expected_delta`.

P&L accounting. At each hourly step t the following quantities are computed and accumulated:

$$\begin{aligned}
\text{Hedge P\&L}_t &= -q_{t-1}^{short}\,(p_t - p_{t-1}),\\
\text{Funding P\&L}_t &= q_{t-1}^{short} \cdot \text{oracle}_t \cdot f_t,\\
\text{Trading fee}_t &= 0.00045 \cdot |\Delta q_t| \cdot p_t \quad \text{(charged at each rebalance)},\\
\text{Net hedge P\&L}_t &= \textstyle\sum_{\tau \le t}(\text{Hedge P\&L}_\tau + \text{Funding P\&L}_\tau - \text{Trading fee}_\tau).
\end{aligned}$$

At each step the following are also recorded:

$$\begin{aligned}
\text{HODL value}_t &= x_0\,p_t + y_0,\\
\text{Gross IL}_t &= \text{HODL value}_t - V_{LP}(p_t),\\
\text{Residual IL}_t &= \text{Gross IL}_t - \text{Net hedge P\&L}_t,\\
\text{Net position P\&L}_t &= \text{LP fee income}_t - \text{Residual IL}_t,
\end{aligned}$$

where LP fee income is the cumulative swap-fee income from Module 4 (forward-filled from daily snapshots). All per-step results are stored in `hedge_results.parquet`.

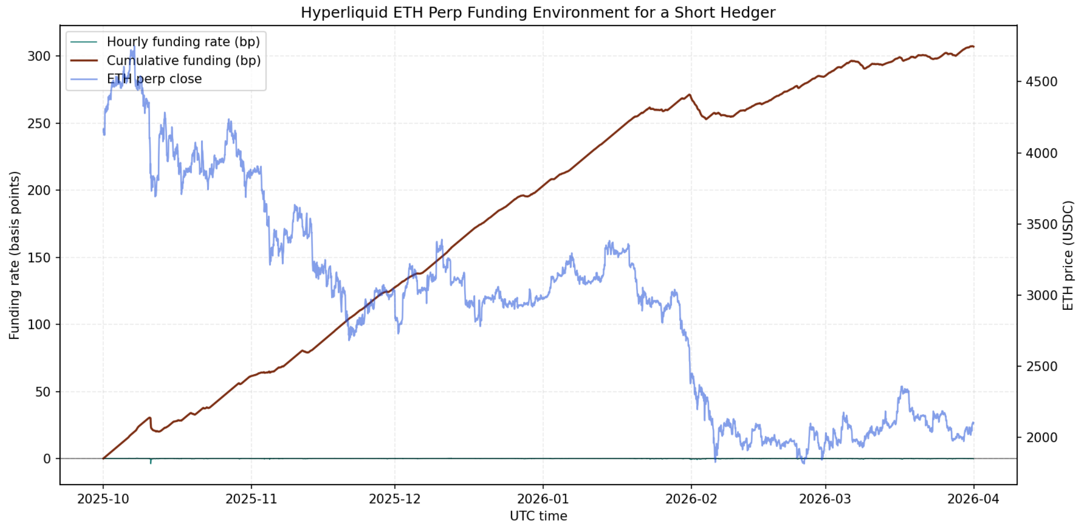

Figure 14: Hourly funding rate and cumulative funding (left axis) vs. ETH perp close price (right axis) over the full study window.

Results. Figure 15 and Table 8 summarise the terminal net LP-plus-hedge P&L for the fixed-frequency schedules. For concentrated positions (P1, P2), the 4h schedule outperforms 1h because fee savings exceed the additional gamma drift accumulated between rebalances. For wider positions (P3, P4), hourly rebalancing limits gamma exposure enough to outperform the coarser schedules on a net basis. P5 (the near-full-range position) is largely insensitive to rebalancing frequency.

Table 8: Terminal net LP-plus-hedge P&L and cumulative trading fees by position and rebalancing frequency (USD).

	1h		4h		24h	
Position	Net P&L	Fees	Net P&L	Fees	Net P&L	Fees
P1	−13 035	967	−9 652	517	−11 205	248
P2	−11 265	994	−9 777	528	−11 007	263
P3	−8 181	754	−10 931	435	−12 421	236
P4	−8 709	854	−10 402	471	−10 360	213
P5	−1 319	220	−1 452	127	−1 317	67

6.4 Active Exit-Recentering Extension

Motivation. The fixed-range LPs in Module 4 are passive positions: each range is entered at the first snapshot and held until the final snapshot. This is a useful benchmark, but it is not a complete model of active market making. Once the ETH price leaves a Uniswap V3 range, the position earns no further swap fees and holds one-sided inventory. A natural active rule is therefore to withdraw and redeploy the same range width around the current price when the old range is out of range.

Strategy design. For each finite-width position P1–P4, the active strategy starts with the same $100,000 deposit and same width k as Module 4. At each hourly Hyperliquid timestamp it marks the LP position, accrues swap-fee income from historical Uniswap swaps whose tick lies inside the current range, and checks whether the hourly close is outside the range. If it is, the strategy withdraws, pays rebalancing costs, swaps into the token ratio required by a fresh range, and redeploys:

$$[p_a, p_b] = [p_t(1-k),\, p_t(1+k)].$$

Costs include the 5 bps Uniswap swap fee on the token notional required to restore the new LP ratio, a $25 gas estimate per recenter, optional slippage (set to zero in the baseline), and Hyperliquid taker fees on hedge adjustments.

HODL-tracking hedge. The active extension also tests a hedge that tracks the original HODL basket rather than merely neutralising the LP's ETH delta. Let E_0 be the initial WETH amount of the Module 4 deposit. The signed long ETH hedge target is

$$q_t = E_0 - \Delta_{LP,t}.$$

Equivalently, under the code's short-positive convention, $q_t^{short} = \Delta_{LP,t} - E_0$. The hedge is recomputed after every hourly mark and after every LP recenter. Performance is measured against the original HODL benchmark:

$$\text{Net}_t = V_{LP,t} + \text{Fees}_t + \text{HedgePnL}_t - V_{HODL,t}.$$

Table 9: Terminal net value versus HODL for passive and active-recentered LP strategies (USD).

Position	Passive	Passive + hedge	Active	Active + hedge
P1	−20 886	−5 209	−33 002	−54 936
P2	−15 203	−1 645	−13 447	−36 968
P3	−16 257	−858	−1 166	−20 584
P4	−14 894	−384	6 277	−4 930

The extension shows why re-entering must be interpreted as an active trading strategy, not as a way to remove IL. Recentering restores fee-earning capacity: active fee income rises to $47.4k for P1, $63.3k for P2, $72.1k for P3, and $38.8k for P4. But the narrow ranges churn heavily. P1 recenters 3,245 times and pays $87.9k in LP rebalancing costs; P2 recenters 1,483 times and pays $41.8k. Wider ranges rebalance less often: P3 recenters 303 times, while P4 recenters only 17 times and is the only active no-hedge strategy to finish above HODL in this baseline. The HODL-tracking hedge helps passive fixed ranges during the downward trend, but it hurts the active recentered strategies here because the repeated inventory resets and hedge adjustments create path-dependent losses. Thus re-entering increases fee income, but it also realizes inventory changes and introduces swap, gas, slippage, and hedge-adjustment costs.

A less reactive threshold rule improves this trade-off. In this variant the LP recenters only when price has moved at least 50% of the range half-width from the current center, a 24-hour cooldown has elapsed, and the trailing 24-hour in-range fee estimate exceeds the estimated rebalance cost. The hedge is also made deliberately lazy with a 5 ETH no-trade band. Under this cost-aware rule, P4 finishes at +$3.1k versus HODL both with and without the HODL-tracking hedge, after 57 recenters. The hedge adds almost no value in this case (net hedge P&L is -$37), but it also no longer destroys the strategy. A sensitivity run with a 75% threshold and 48-hour cooldown gives the strongest unhedged result, P2 at +$24.0k versus HODL and P4 at +$7.2k, but the hedged versions remain weaker.

6.5 Limitations and Critical Discussion (Task 5.4)

Residual gamma risk. A short perpetual cancels today's delta but does not remove curvature. Between rebalances the hedge drifts from the true delta by an amount driven by Γ_{LP}; the resulting convexity loss is approximated by $\frac{1}{2}\Gamma_{LP}(p_t)(\Delta p_t)^2$ accumulated over all hours. This proxy totals −$2,496 (P1), −$7,723 (P2), −$9,521 (P3), −$9,845 (P4), and −$2,444 (P5). Narrow positions carry the highest gamma exposure per dollar because they require larger L; infrequent rebalancing compounds this by allowing larger inter-period price moves.

Funding-rate risk. Over the study window the mean hourly funding rate is 7.3×10^{-6}, providing the short hedger a small persistent credit. During the 219 large-move hours (absolute log return above the 95th-percentile threshold of 1.52%), the mean rate drops to 1.7×10^{-6}: funding income is smallest precisely when gamma losses are largest. This adverse correlation means funding receipts do not reliably cushion the worst impermanent-loss events.

Minimum viable position size. Cumulative trading fees under the 1h schedule range from \$220 (P5) to \$994 (P2) over six months. An LP must earn enough in swap fees to cover these costs plus any residual IL. Positions whose expected fee income falls short of this threshold cannot justify continuous 1h rebalancing; switching to 4h or a no-trade-band schedule reduces the fee burden by 40–50%, lowering the break-even notional accordingly.

Options as an alternative hedge. A perpetual hedges delta but not gamma. A short-dated ETH put spread or straddle with strikes at the position's range boundaries would directly offset the concavity left unhedged by the perpetual, eliminating residual gamma risk entirely. The key assumptions required are sufficient options liquidity at the relevant strikes and tenors, and that implied volatility at the time of purchase is below the realised volatility over the hedge horizon, otherwise the option premium exceeds the gamma loss it prevents. With annualised realised volatility at 69% over this study window, options merit serious consideration for the most concentrated positions, P1 and P2, where the gamma proxy exceeds \$2,500 and \$7,700 respectively.

7 Conclusion

This report reconstructs the Uniswap V3 USDC/WETH 0.05% pool over the period from 1st of October 2025 to 31st of March 2026 and uses that reconstruction to connect three quantities that are often studied separately: the location of liquidity, taker execution cost, and liquidity-provider performance. The dataset covers more than one million swaps, roughly \$21.4 billion in USDC notional, hundreds of thousands of liquidity-changing events, and 182 daily pool-state snapshots. The validation checks against slot0 prices and sampled on-chain tick states give confidence that the reconstructed pool state is accurate enough for the downstream empirical analysis.

The central result is that liquidity depth is spatial, not merely aggregate. As ETH fell from roughly \$4,150 to \$2,025, liquidity migrated toward the realised trading range, with a persistent depth band around \$2,000. The $\pm 5\%$ in-range liquidity ratio rose from 5.0% to 36.1%, while the global liquidity HHI remained only weakly correlated with price. This distinction matters: TVL measures how much capital is deposited in the pool, but execution quality depends on whether that capital is close to the marginal price.

The taker-side analysis makes this mechanism visible. Against reconstructed daily pool states, small trades remain close to the 5 bps fee floor, while \$1M trades face median simulated price impact of about 102 bps and much larger state-dependent dispersion. Observed effective spreads follow the same increasing pattern with trade size, but they also reflect routing, within-block ordering, MEV, and strategic order splitting. The simulator should therefore be interpreted as a clean measure of single-pool mechanical depth, not as a complete model of realised execution.

For liquidity providers, the same concentration mechanism creates a sharp inventory-risk trade-off. Narrow ranges earn more fees per dollar while active, but they become more exposed to large directional moves and to the short-gamma profile of concentrated liquidity. Across the five representative \$100,000 positions, cumulative fees did not offset impermanent loss while every position ended with negative net P&L. The full-range position earned the least in fees but lost the least overall, while concentrated positions earned more fees yet suffered materially larger IL. Delta hedging with ETH perpetuals reduced residual exposure but did not restore profitability, because discrete hedges leave gamma risk, incur trading costs, and receive the least funding support during large-move periods. The active-recentering extension confirms the same trade-off from another angle: re-entering can restore fee-earning capacity, but narrow ranges churn so often that swap, gas, and hedge-adjustment costs can overwhelm the additional fees.

Several limitations bound these conclusions. The swap simulator uses daily snapshots and abstracts from gas, multi-hop routing, private order flow, and same-block strategic behaviour. The observed effective-spread comparison is based on a stratified block sample rather than every swap block. The LP analysis is counterfactual: synthetic liquidity is included in the fee denominator, but the historical swap path is held fixed. The hedging backtest also uses simplified trading costs and discrete rebalancing. The active-recentering extension uses an hourly rule and an event-level fee proxy rather than a full endogenous pool simulation. These choices make the results interpretable and reproducible, but not a full equilibrium model of how the pool would have evolved under different liquidity provision.

Overall, the study shows that concentrated liquidity cannot be evaluated by TVL alone. During this window, liquidity migration improved nearby execution for takers as the market moved lower, but LPs positioned around the prior price regime were not compensated enough for the resulting directional and short-gamma exposure. For takers, concentrated liquidity can make execution cheap when capital follows the price; for LPs, the same mechanism turns liquidity provision into a path-dependent volatility-selling strategy whose fees may be insufficient in trending markets.

A Appendix

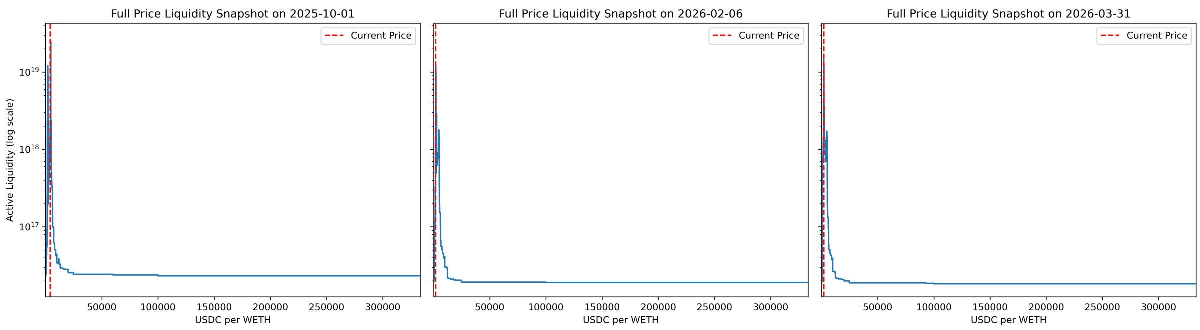

Figure 17: Fig 2.1 — Full price-range liquidity profiles for the Module 2 representative snapshots.

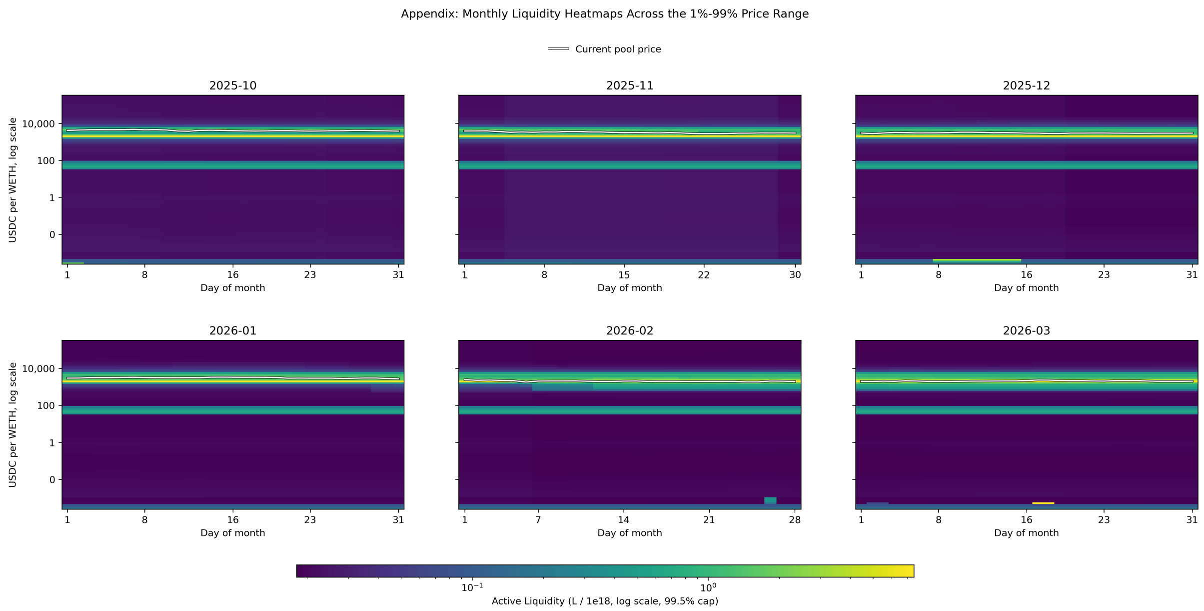

Figure 18: Fig 2.1b — Monthly liquidity heatmaps across the broad price range. The vertical axis is on a log price scale and excludes the extreme 1% tails of initialized ticks so the economically relevant range remains visible.

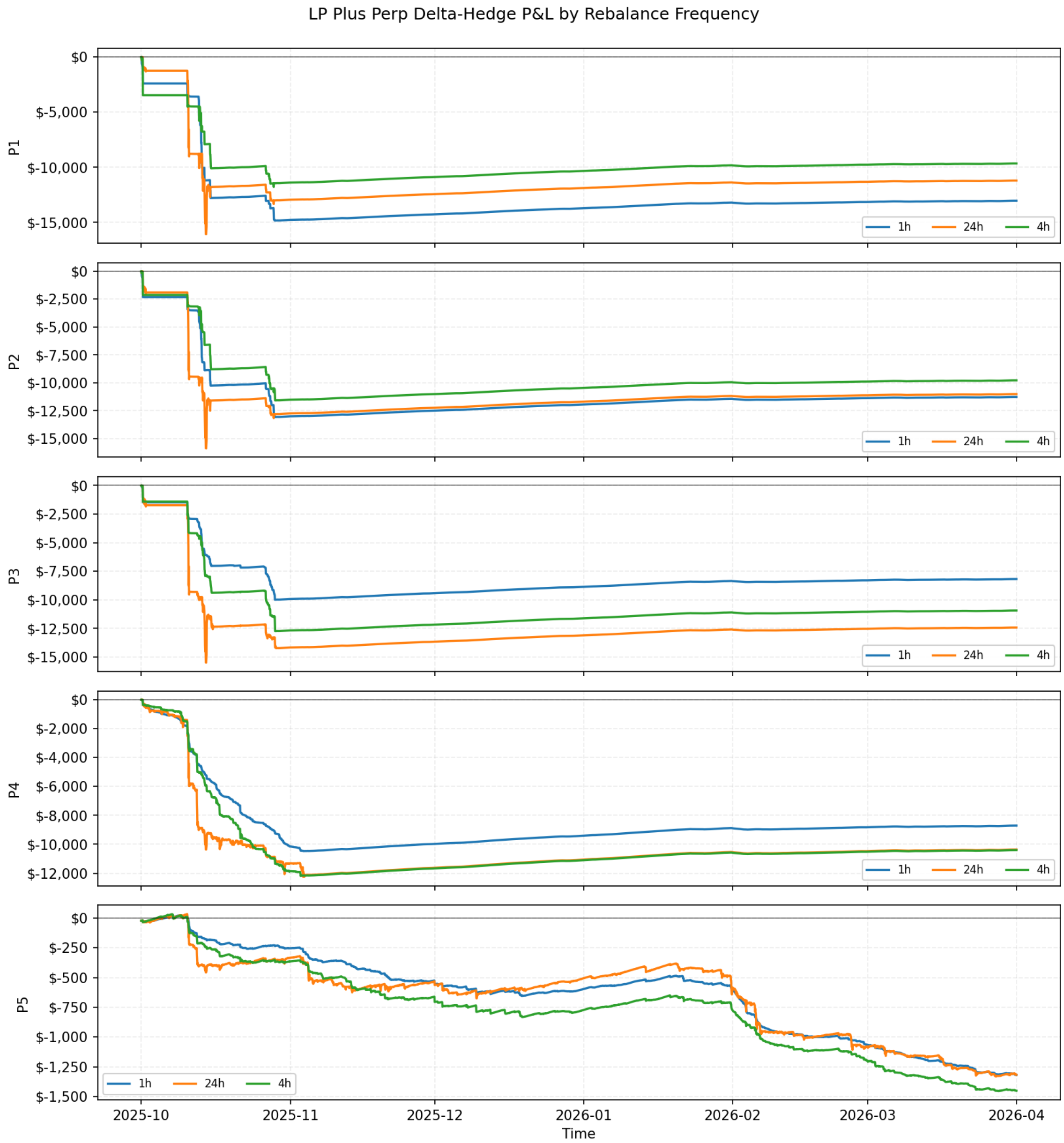

Figure 15: Cumulative net LP-plus-hedge P&L by position and rebalancing frequency. Hedge P&L, funding income, and trading costs are all included.

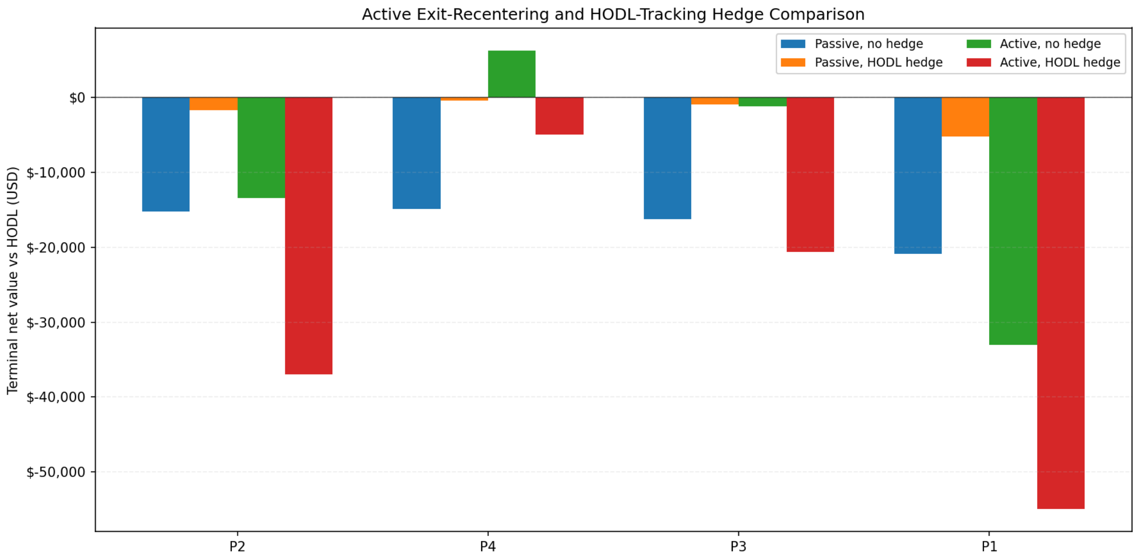

Figure 16: Terminal net value versus the original HODL basket for passive fixed-range LPs and active exit-recentered LPs, with and without the HODL-tracking hedge.